Plant foods and the risk of cerebrovascular diseases: a potential protection of fruit consumption.

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Studies on the association between plant foods and cerebrovascular diseases have given contradictory results suggesting the existence of some effect-modifying factors. The present study determines whether the consumption of plant foods (i.e. fruits and berries, vegetables, and cereals) predicts a decreased cerebrovascular disease incidence in a population with low fruit and vegetable and high wholegrain intake. This cohort study on 3932 men and women was based on data from the Finnish Mobile Clinic Health Examination Survey, conducted in 1968-72. The participants were 40-74 years of age and free of cardiovascular diseases at baseline. Data on the plant food consumption were derived from a 1-year dietary history interview. During a 24-year follow-up 625 cases of cerebrovascular diseases occurred, leading to either hospitalisation or death. An inverse association was found between fruit consumption and the incidence of cerebrovascular diseases, ischaemic stroke and intracerebral haemorrhage. The adjusted relative risks (RR) between the highest and lowest quartiles of intake of any cerebrovascular disease, ischaemic stroke and intracerebral haemorrhage were 0.75 (95 % CI 0.59, 0.94), 0.73 (95 % CI 0.54, 1.00) and 0.47 (95 % CI 0.24, 0.92), respectively. These associations were primarily due to the consumption of citrus fruits and occurred only in men. Total consumption of vegetables or cereals was not associated with the cerebrovascular disease incidence. The consumption of cruciferous vegetables, however, predicted a reduced risk of cerebrovascular diseases (RR 0.79; 95 % CI 0.63, 0.99), ischaemic stroke (RR 0.67; 95 % CI 0.49, 0.92) and intracerebral haemorrhage (RR 0.49; 95 % CI 0.25, 0.98). In conclusion, the consumption of fruits, especially citrus, and cruciferous vegetables may protect against cerebrovascular diseases.